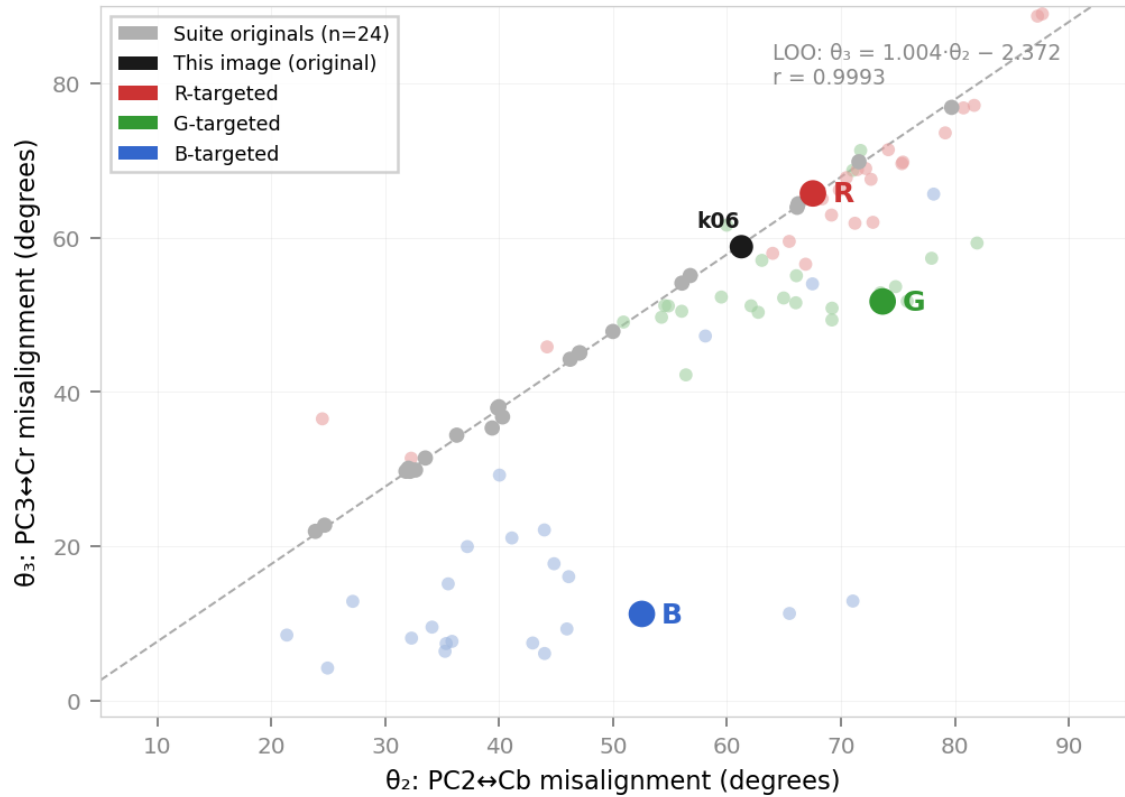


Compression Profile: KODIM06 768x512 | 24-bit RGB | PNG (lossless)

sailboat at anchor | Near-Degenerate (PC1 = 98.42%)

1. Geometric Displacement Under Directional Perturbation



2. Pairwise Correlation and Angular Misalignment

Version	R-G	R-B	G-B	Avg r	theta2	theta3	L00 dev
Original	0.9772	0.9603	0.9911	0.976	61.26°	58.85°	-0.28°
R-targeted	0.1377	0.1006	0.9539	0.397	67.54°	65.76°	+0.32°
G-targeted	-0.1464	0.9723	-0.1225	0.414	73.66°	51.75°	-19.83°
B-targeted	0.9605	-0.1457	-0.2880	0.465	52.52°	11.23°	-39.13°

R-targeted isolates red (R-G, R-B < 0.14) while preserving G-B coupling at 0.954.

G-targeted isolates green (R-G, G-B < 0.15) while preserving R-B at 0.972.

B-targeted preserves R-G at 0.960 while driving R-B and G-B near zero.

All three perturbation directions depart from the L00 constraint manifold ($r = 0.9993$).

Compression Profile: KODIM06768x512 | 24-bit
RGB | PNG (lossless)

sailboat at anchor | Near-Degenerate (PC1 = 98.42%)

3. File Format and Size Comparison

All versions depict the same scene. BPP = (file size in bits) / (pixel count).

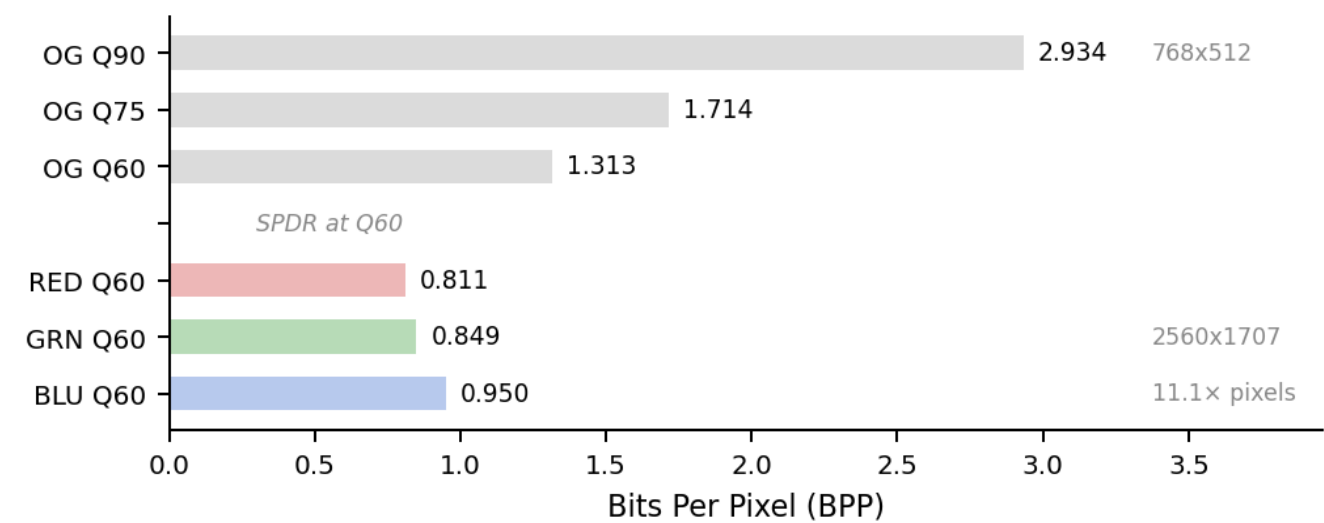
ORIGINAL (kodim06)

Version	Format	Resolution	Size	BPP	Pixels
PNG (lossless)	PNG	768x512	604.5 KB	12.593	393,216
JPEG Q90	JPEG	768x512	140.8 KB	2.934	393,216
JPEG Q75	JPEG	768x512	82.3 KB	1.714	393,216
JPEG Q60	JPEG	768x512	63.0 KB	1.313	393,216

REDISTRIBUTED (kodim06 SPDR at Q60)

Version	Format	Resolution	Size	BPP	Pixels
RED Q60	JPEG	2560x1707	432.4 KB	0.811	4,369,920
GRN Q60	JPEG	2560x1707	453.0 KB	0.849	4,369,920
BLU Q60	JPEG	2560x1707	507.0 KB	0.950	4,369,920

4. Quality Ladder: SPDR Q60 vs Original Q90



All three perturbation directions at Q60 produced lower BPP than the original at Q90, while carrying 11.1x the original pixel count.

Compression Profile: KODIM06768x512 | 24-bit RGB | PNG (lossless)

sailboat at anchor | Near-Degenerate (PC1 = 98.42%)

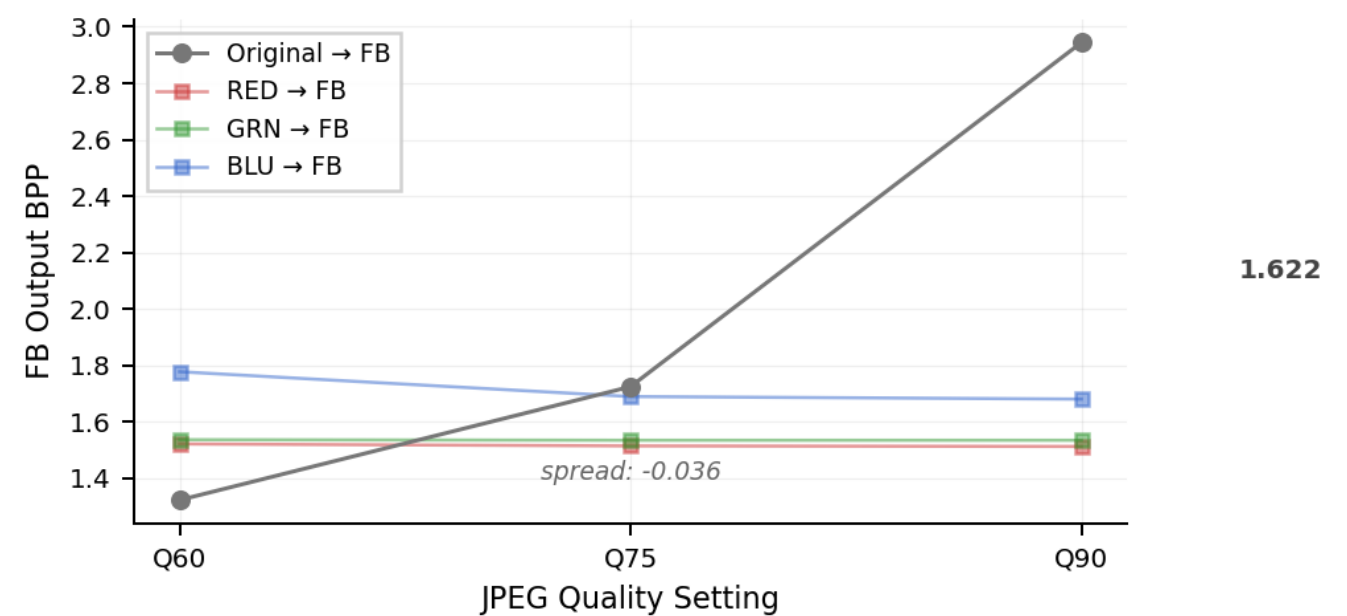
5. Third-Party Compression Pipeline (Facebook/Meta)

Each version uploaded to Facebook. Platform-compressed output measured at steady state (FB2).

Pipeline	Input Res	Output Res	FB BPP	BPP reduction
OG PNG → FB	768x512	768x512	2.795	–
R-targeted → FB	2560x1707	2048x1366	1.512	45.9%
G-targeted → FB	2560x1707	2048x1366	1.534	45.1%
B-targeted → FB	2560x1707	2048x1366	1.680	39.9%
ABC (no perturbation) → FB	2560x1707	2048x1366	1.433	48.7%

ABC control: same pipeline, same resolution, no covariance perturbation. Avg |r| unchanged (0.9762 vs 0.976 original). Comparable BPP reduction was not observed in the ABC control.

6. Quality Parameter Sensitivity Through Facebook



Original: Q60 → Q90 produces a 1.622 BPP spread through Facebook.
Redistributed: Q60 → Q90 produces a 0.036 BPP spread through Facebook.
Redistributed versions produced substantially smaller Q60-90 BPP variation through the measured pipeline.

7. Pipeline Summary – kodim06

Best BPP (Q60, direct)	0.811	2560x1707, 4,369,920 px
Best BPP through FB	1.512	2048x1366 output
vs OG Q90 → FB	48.6%	SPDR Q60→FB vs OG Q90→FB
Pixel ratio	11.1×	4,369,920 vs 393,216